

FX Nexus: Forex Data Engine

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Course Project Report

Database Systems Course

Project Context

FX Nexus is a database system designed to store, analyze, and backtest Forex market data.

The system provides a centralized platform where traders can efficiently query historical data, compute technical indicators, and simulate strategies. It serves both as an academic tool for database learning and a practical platform for strategy analysis in Forex trading.

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1 Introduction

The foreign exchange (Forex) market is the world's largest financial market, with daily trading volumes on the order of trillions of dollars. In fact, a recent survey reported average daily FX turnover of about US\$9.6 trillion (April 2025) . This market operates 24/5 and involves participants worldwide (banks, funds, corporations, etc.), creating massive amounts of time-stamped market data every second. Managing and analyzing this high-frequency data is crucial for developing and validating trading strategies.

In an academic context, a Forex trading database provides a practical learning platform for database and data analysis courses. Students can experience real-world data challenges: designing scalable schemas, importing diverse data sources, indexing time-series data, and performing SQL-based analytics. A well-designed database allows fast retrieval of historical price series and embedded computation of technical indicators, making it easier to test hypotheses and strategies efficiently.

2 Problem Statement

Existing tools for Forex analysis have several limitations:

- **Fragmented Data Sources:** Forex price data comes from many vendors (brokers, news feeds, data providers, etc.), each with its own format and coverage. The market itself is fragmented among many trading venues, making it difficult to obtain a comprehensive, unified dataset.
- **Error-Prone Manual Analysis:** Analysts using spreadsheets or standalone scripts are prone to errors and inconsistencies, leading to potentially wrong conclusions or missed opportunities.
- **Performance and Scalability Constraints:** High-frequency strategy backtesting requires rapid access to vast datasets, which traditional systems often cannot handle efficiently.

FX Nexus addresses these problems with a centralized, indexed database, built-in SQL analytics, and fast query performance.

3 Objectives

- Scalable Schema Design for high-frequency Forex data.
- Multi-Source Data Ingestion for unified dataset.
- Efficient Analysis Tools (MA, RSI, MACD).
- Backtesting Capability to simulate strategies.
- Fast Query Performance with optimized indexing.

4 Scope

- Historical Price Storage (M1, M15, H1)
- Source Tracking
- Technical Indicators
- Backtesting Reports

5 Data Collection

- Financial APIs: Yahoo Finance, OANDA.
- Trading Platform Exports: MetaTrader 4/5.
- Other sources: Kaggle datasets, vendor packages.

6 Technology Stack

- DBMS: MySQL, PostgreSQL, TimescaleDB.
- Programming: C++, Python.
- SQL/Analytics Tools: pgAdmin, MySQL Workbench.
- Additional Tools: Git, Docker, visualization libraries.

7 System Functionality

1. Data Ingestion
2. Data Storage and Indexing
3. Price Retrieval Queries
4. Technical Indicator Calculation
5. Backtesting Module
6. Reporting and Export

8 User Classes

All users are traders/analysts with equal permissions. Users can:

- Access the database securely.
- Import market data.
- Execute SQL queries and run backtests.
- Export results.

9 Conclusion

FX Nexus provides a centralized and efficient database system for managing, analyzing, and backtesting Forex market data. The system addresses fragmented data sources, limited analytical support, and slow access. Expected benefits:

- Faster access to historical Forex data.
- Improved strategy evaluation (MA, RSI, Breakout).
- Educational benefits for database and finance courses.
- Scalable design for future expansion.

10 References

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